

Name: Ninja Cat

Date: 11/11

Element B: Documentation and Analysis of Prior Design Attempts

From our interview with the client, we know that there have been no prior design attempts for his cart specifically. However, we will attempt to draw parallels between other design solutions, such as for pushing a stroller, wheelchair, or other wheeled objects with handles.



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Prior Design Attempt #1 - Wagon-Mode Push Handlebar

larktale.com/products

Figure 1.1 - Larktale's push attachment for their crossover stroller



Details

Pros	Cons
• Adjustable length side bars (no dimensions given, Larktale) • Leatherette grip on handle for comfort (Larktale) • Can fold down at joint to save space (Larktale) • ~45-degree pushing angle for efficient use of pushing force • Easy removal using push buttons (Larktale)	• Extremely limited vertical freedom to change height • Designed specifically for Larktale's own stroller (Larktale)

Analysis

The sales page for this product showed 0 reviews (Larktale), indicating that it wasn't widely bought. However, I believe that this was due to the niche nature of the product. Overall, the design seems to be well thought out, especially with the ability to fold down for storage, ease of removability, and comfortable grip. However, the angle of the device means taller users may still have to bend to use it comfortably.

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Prior Design Attempt #2 - Folding Wagon Pushbar

<https://whitsundayshop.com/products/>

Fig 2.1 - Whitsunday's folding wagon push bar



Details

Pros	Cons
Can fold down at the hinge	- Unable to change height - Tools are required to move from handle to handle - Only clamps on one bar size - Basic/unergonomic handle

Analysis

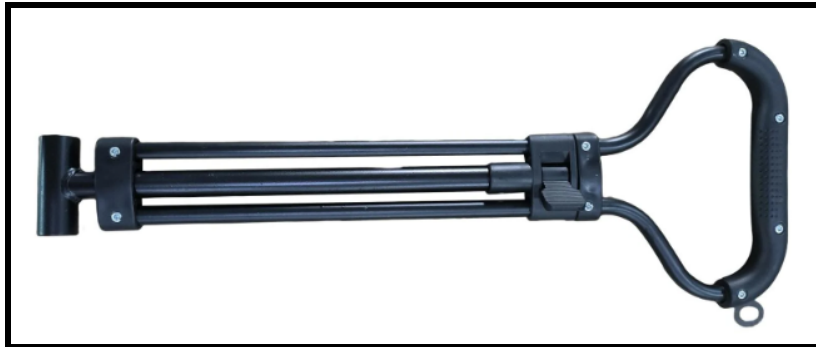
This solution is extremely rudimentary, lacking a lot of the features of PDA #1. However, it seems to still accomplish its goals, as seen by the [insert reviews] (Whitsunday). The fundamental structure of the design could be further improved by making the main arm extendable and improving the design of the handle to be more ergonomic.

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Prior Design Attempt #3 - Pull Handle Accessory

<https://www.walmart.com/ip/Sharplace-Pull-Handle-Accessory-Parts-Collapsible-Wagon-Cart-Black-Park/1093128195?classType=REGULAR&athbdg=L1700>

Fig 3.1 - Walmart's pull handle accessory



Details

Pros	Cons
• Can attach to a couple of different sizes of bar • Easy to attach and remove with lever mech (Walmart) • Rounded and rubber handle, somewhat ergonomic • Extendable (Walmart)	• Only for pulling (Walmart) • May still have issues for taller users, as the clamp is still more designed for a bracing bar instead of an existing handle

Analysis

This design is rather simple, elegant, and seems to accomplish many of our goals, especially being able to be removed without tools, being extendable, and having a nice handle. However, the fact that it's only designed for pulling is a serious drawback for our situation.

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Prior Design Attempt #4 - Electric Office Wheelchair

US Patent Office -

<https://ppubs.uspto.gov/dirsearch-public/print/downloadBasicPdf/20240216192>

Fig 4.1 - Full Device Exploded View

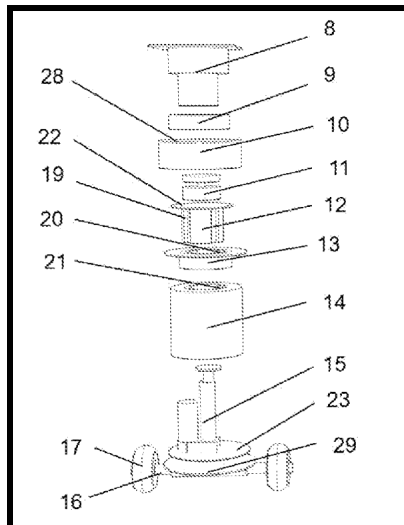
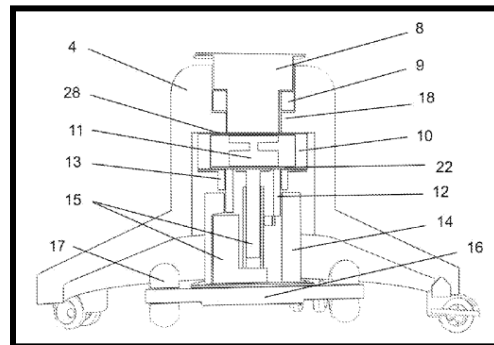


Fig 4.2 - Electronic Base Exploded View



[0015] Reference numerals: (1) a chair back, (2) armrests, (3) a seat, (4) a chassis, (5) a joystick, (6) a battery and control module, (7) casters, (8) a support, (9) a plane bearing, (10) a cover tube, (11) a pressure sensor, (12) a telescopic tube, (13) a bottom tube, (14) a column tube, (15) an electric pusher, (16) a bridge frame, (17) electric wheels, (18) a platform, (19) chutes, (20) bottom slide-ways, (21) slide-ways, (23) a bottom plate, (24) a pressure sensor module, (25) a telescopic device, (26) a driving device, (27) limit switches, (28) a cover plate, and (29) a bearing plate.

Details

While this "solution" solves a slightly different problem than we have set out to, it represents the idea of a motorized solution.

Pros	Cons
- Motorized -> no need to push at all - Possibility for autonomous following - Allows user to move faster than they would pushing	- Needs to be charged - More expensive than a purely mechanical solution - Harder to repair

Analysis

This design represents a more complex solution, which would require more time to develop, more budget, and access to parts not currently available. However, this solution has the ability to completely eliminate any back pain from the user, providing the ultimate ergonomics.

